



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

RESIT EXAMINATION JANUARY 2019

Course : EC3239 (Operating Systems Principles)

Time : 1:00pm – 4:00pm
(3 hours)

Lecturer : Wan Nor Al-Ashekin Binti Wan Husin

Date : 16 December 2019

Instructions:

Answer ALL questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 4 printed pages
(excluding front cover).

Cheating is a very serious offence and those caught will be severely punished
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

Question 1

[25 MARKS]

a) Explain the responsibilities of following operating system managers:

- Device manager
- Processor manager

(6 marks)

b) Describe the following types of operating system:

- Real-time systems
- Embedded systems

(4 marks)

c) Explain the shared memory model in inter-process communication.

(4 marks)

d) Consider the information in Table 1:

Table 1

Process Name	Burst Time
P1	12
P2	15
P3	22
P4	12

i) Show the Gantt chart to illustrate the execution of the processes in Table 1 using Round Robin Scheduling algorithm with the time quantum = 10,

(5 marks)

ii) Write the turnaround time of the processes in Table 1 using Round Robin scheduling algorithm.

(2 marks)

e) Explain the deadlocks situation in dedicated device allocation.

(4 marks)

Question 2

[25 MARKS]

- a) Explain fix memory partition and variable memory partition.
(6 marks)
- b) Assume that the main memory has the following five fixed partitions with the following sizes: 50KB, 250KB, 150KB, 450KB and 750KB (in order).

Show how the First-fit algorithm would place the processes 210KB, 400KB, 140KB and 650KB (in order).
(6 marks)
- c) Describe file system in operating system. Include file concept, file attributes and file operations in your description.
(8 marks)
- d) Explain the central processing unit issue in operating system.
(5 marks)

Question 3

[25 MARKS]

- a) Model the relationship of non-network operating system with its resources category.
(10 marks)
- b) The distributed systems work in Wide Area Network (WAN) that involved more communication between nodes as compared to computation. The users have an impression that they are working on single machine. Explain the communication in distributed systems.
(6 marks)
- c) Differentiate THREE (3) issues in distributed operating systems.
(9 marks)

Question 4

[25 MARKS]

a) Consider the following file structure (Figure 1):

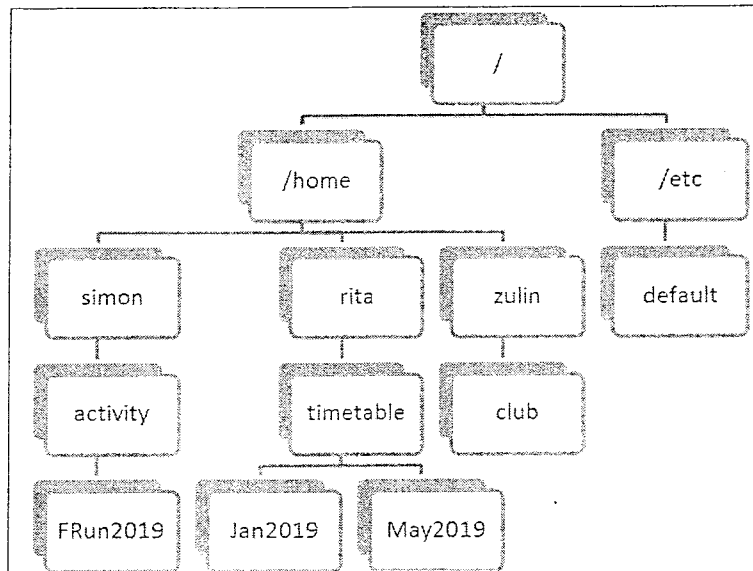


Figure 1

Write an appropriate Linux command for the following statements:

- View all files in directory `timetable` from directory `/etc`.
- Change directory from directory `/etc` to directory `/FRun2019`.
- Change directory from directory `/FRun2019` to directory `activity`.
- View content of file `default` from directory `activity`.
- Change directory from `/etc` to directory `zulin`.

(5 marks)

b) Write an appropriate Linux command for the following statements:

- Adira has a file called `/home/adira/exercise` and she wants to copy it to `/tmp` with the file name, `class_exercise`.
- Nabilla wants to move a file named `resource.txt` from syura's home directory to the next lower directory, `my_agenda`.
- Naveen wants to remove a file called `EC3239_NameList`.
- Misbun wants to create the directories `Sport`, `Healthy_info` and `Fans` in the home directory.
- You have a file called `member.conf` and you want to create a hard link called `hardlinkmember.conf`.
- You would like to add a new group called `RecreationClub`.
- Jasmine wants to add a new user called `Mawar` from system.
- Jessica wants to modify a user ID of user Brown from 1223 to 1443.
- Marvin wants to switch account to Farid's user account.
- You want to display all files in directory `/sbin` from your current working directory, `EC2301`.
- Juita wants to move file `ingredients` from directory `/home/juita` to directory `/tmp`.

(12 marks)

c) Consider the information given in Figure 1:

```
[SuSE tmp 1298]$ ls -l
total 32
-rw-rw---- 1 firdaus 1201 1523 2019-04-11 11:56 assignment1.txt
-rwxrw---- 1 firdaus 1201 2121 2019-03-12 11:56 quiz
-rw-r--r-- 1 firdaus 1201 2345 2019-03-10 11:56 convertNumber.c
-rw----- 1 firdaus 1201 10760 2019-03-01 11:50 conf.
-rw----- 1 firdaus 1201 1020 2019-03-01 10:50 classwork.sh
```

Figure 1

Write a single Linux command you would use to accomplish each of the following goals, in a minimal fashion. There should be no other rights granted. Assume the commands will be entered in the current working directory whose contents are shown above:

- i) Allow the owner to execute the file *assignment1.txt*.
- ii) Prevent all users except the owner from view and modify file *quiz*.
- iii) Allow all members of group 1201 to modify the contents of file *convertNumber.c*.
- iv) Base on permission in (iii), allow other members to modify and execute the file *convertNumber.c*.
- v) Allow the owner to execute, group and others to read the file *conf*.
- vi) Base on permission in (v), allow the group members to modify the group and other users have no access to file *conf*.
- vii) Change ownership of file *classwork.sh* from firdaus to janna.
- viii) Allow all members of group 1201 to view the contents of file *classwork.sh*.

(8 marks)

-THE END -